

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree Examination December 2022 (2019 scheme)

Course Code: EET435**Course Name: RENEWABLE ENERGY SYSTEMS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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| 1 | Explain about the role of non-conventional energy sources on the basis of current energy demands of the world. | (3) |
| 2 | Explain the principle of solar energy conversion into heat energy. | (3) |
| 3 | List out any four environmental impacts of OTEC systems | (3) |
| 4 | What are the different components of the tidal power plant? Explain. | (3) |
| 5 | Three-bladed wind rotor with blade length of 52 m is operating in a wind stream having wind velocity of 13 m/s. Air density is 1.23 kg/m^3 and power coefficient may be taken as 0.3. Calculate the extractable power from the wind. | (3) |
| 6 | Discuss the advantages and disadvantages of horizontal and vertical axis wind mills. | (3) |
| 7 | Define biomass gasification. List different types of biomass gasifiers. | (3) |
| 8 | What are the factors affecting the biogas generation? | (3) |
| 9 | What are the risks for the hydrogen energy storage? | (3) |
| 10 | What are fuel cells? List out the applications of fuel cells. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | a) What are solar collectors? Give their classification and compare them based on construction and area of applications. | (10) |
| | b) Write about any three types of solar cells based on construction materials | (4) |

OR

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| 12 | a) Explain about stand alone and grid connected solar photo voltaic systems | (10) |
| | b) Explain the voltage- current characteristics of a solar cell with neat sketch | (4) |

Module II

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| 13 | a) Discuss the basic principle of OTEC. Describe a closed cycle OTEC with its advantages and disadvantages. | (10) |
| | b) What is biofouling? How it affects efficiency of energy conversion and how can it be minimised? | (4) |

OR

- 14 a) What are the different classifications of tidal power plants. Explain the components and detailed working operation of double basin tidal power plant with neat sketch. (10)
- b) What are the limitations of the tidal power production? (4)

Module III

- 15 a) What is mean by betz limit? Derive the expression for the power extracted by a wind turbine. (8)
- b) What are the different types of generators used with wind turbines? (6)

OR

- 16 a) Describe the construction of a three-bladed horizontal shaft wind turbine generator unit. Explain the terms yaw control and pitch control. (8)
- b) What are the advantages and disadvantages of a vertical axis wind turbine system? (6)

Module IV

- 17 a) Describe the construction and working of a biogas plant, its material aspects, and utilization of plant products with a neat diagram (10)
- b) Explain the advantages and uses of biogas. (4)

OR

- 18 a) What are the different types of biogas plants? Explain the construction, working of a fixed dome type biogas plant and floating dome type biogas plant (10)
- b) What are the factors affecting the selection of a particular model of a biogas plant? (4)

Module V

- 19 a) Explain about the components of a small hydro electric power generation scheme with the help of a neat diagram. (8)
- b) Explain about any two methods of hydrogen production. (6)

OR

- 20 a) Explain the working of a H_2O_2 fuel cell and also write the advantages and disadvantages of a fuel cell (8)
- b) What are the advantages and applications of hydrogen energy? (6)
